

ALI AKGÖZ

PROJECT MANAGER & R&D SPECIALIST

Radioactive Waste Management | Nuclear Safety | Technical R&D | International Project Delivery

Ankara, Türkiye | +90 537 277 45 87 | aliakgoz@gmail.com | aliakgoz.github.io

PROFESSIONAL SUMMARY

Civil engineer, project manager and R&D specialist with 15+ years of experience across nuclear safety, radioactive waste management, regulated infrastructure and software-enabled research. Leads Türkiye's first near-surface radioactive waste disposal facility project and manages an EU-funded international cooperation programme, translating policy, regulatory and engineering objectives into clear scope, workstreams, milestones, dependencies, acceptance criteria and coordinated execution. Experienced in technical specifications, public procurement, management systems, quality assurance and international stakeholder coordination. Also develops sensor-to-cloud, mobile, web, Python and machine-learning applications for radiation monitoring and scientific analysis.

CORE EXPERTISE

Project & Program Management | Radioactive Waste Management | Nuclear Safety & Licensing | International Stakeholder Coordination | Technical Specifications & Procurement | Requirements, Milestones & Acceptance Criteria | Management Systems & QA/QC | Risk & Issue Management | R&D & Systems Integration | Python, Data & Machine Learning

PROFESSIONAL EXPERIENCE

TENMAK - Turkish Energy, Nuclear and Mineral Research Agency | Ankara, Türkiye | May 2019-Present

R&D Specialist / Project Manager, Radioactive Waste Management Coordination Department

Project Lead - Türkiye's First Near-Surface Disposal Facility

- Lead the program to develop Türkiye's first near-surface radioactive waste disposal facility, coordinating site characterization, facility design, safety case / safety analysis report, waste acceptance criteria, management-system, quality, licensing and stakeholder workstreams.
- Own day-to-day execution across interdependent technical and procurement workstreams; convert high-level business, policy, regulatory and engineering objectives into roadmaps, milestones, dependencies, review gates and acceptance criteria, while making scope and sequencing trade-offs to protect the critical path.
- Co-lead preparation and technical review of a ~650-page procurement specification covering repository design and preparation of the safety analysis report / safety case, with structured requirements, traceability, QA/QC, document control, evaluation logic and objective acceptance provisions.

Project Manager - EU-Funded INSC Cooperation Program

- Manage the EU-funded Instrument for Nuclear Safety Cooperation (INSC) project, 'Cooperation with the Turkish Organisation in Charge of Radioactive Waste Management' (TR4.01/23; Contract No. INSC/2024/459-887), aligning TENMAK, European Commission and EU stakeholders, international contractors and multidisciplinary technical teams.
- Lead progress reviews, scope clarification, action and decision tracking, issue/risk escalation and deliverable review; translate complex technical trade-offs into decision-ready communication for senior management, government, regulatory and international project governance.

Technical Procurement, Quality & R&D Delivery

- Led preparation of the technical specification for new radioactive-waste storage facilities in Istanbul and supported the resulting public procurement, including market sounding, supplier outreach, quotation collection, tender documentation and technical clarification in a highly specialized supplier market.
- Apply management-system, organizational-readiness and quality-management principles to technical delivery, including roles and responsibilities, configuration and document control, auditability, nonconformance and corrective-action workflows, records management, supplier quality and final acceptance.
- Standardize reusable delivery patterns - requirement coding, traceability matrices, review workflows, acceptance evidence and document-control templates - while synthesizing international standards, regulatory expectations and operating experience into actionable requirements for national radioactive-waste planning.
- Moved radiation-monitoring concepts from 0-to-1 working prototypes, integrating detector hardware, Python analytics, cross-platform mobile and web applications, cloud data storage, wireless interfaces and geospatial visualization.

Turkish Atomic Energy Authority (TAEK) | Ankara, Türkiye | Apr 2014-May 2019

Atomic Energy Specialist / Nuclear Safety Inspector

- Supported licensing of the Akkuyu Nuclear Power Plant through review and assessment of civil and structural sections of the Preliminary Safety Analysis Report and associated engineering submissions.
- Performed nuclear-construction and quality-management inspections on behalf of the national regulatory authority; documented findings, evaluated corrective actions and coordinated technical interfaces with licensees, designers and contractors.
- Co-authored national regulations on nuclear power plant construction inspection and management systems in nuclear installations, translating IAEA requirements into implementable provisions and supporting organizational readiness for adoption.
- Completed an accepted professional expertise thesis on soil-structure interaction modelling of nuclear power plant structures, developing 3D finite-element and stick models and performing SASSI analyses.

EARLIER ENGINEERING EXPERIENCE

Koltek Consultancy | Ankara, Türkiye | 2012-2014

Structural Engineer

- Designed and analyzed industrial structures, reviewed contractor submissions and served as control engineer for design compliance and construction quality.
- Coordinated clients, engineering disciplines and site teams to resolve design, constructability and execution issues.

Eser-Onur-Nesma Partnership | Jeddah, Saudi Arabia | 2011-2012

Technical Office Engineer - Jeddah Stormwater Collection Project

- Delivered technical-office design and analysis support and performed construction-site quality control for a major stormwater infrastructure program.

SELECTED TECHNICAL & DIGITAL WORK

AI-Assisted Technical Workflows - Early OpenAI Playground adopter from the pre-ChatGPT period; currently uses ChatGPT Pro and other tools for requirements decomposition, technical-document comparison, specialised market research, bilingual drafting and decision support, with expert review and traceability.

Radiation Monitoring Platform - Designed an end-to-end sensor-to-cloud architecture combining a handheld detector, Bluetooth connectivity, a Flutter mobile application, cloud database and web dashboard for processing, visualization and data archiving.

UAV Radiation Mapping - Developed an RF-connected web application to display live sensor readings on maps and apply kriging-based geospatial interpolation; also developed machine-learning and signal-processing workflows for radiation-related R&D.

Scientific Analytics Modernization - Ported a complex PCA/SVD-based scientific workflow from R to Python and improved execution speed by approximately 3x using NumPy, SciPy, pandas, HDF5 and data-visualization libraries.

SELECTED INTERNATIONAL EXPERT ASSIGNMENTS

- **OJT 2.2.17-1 - Design of Radioactive Waste Management Facilities** | Finland | 4 May-12 Jun 2026 | 35-day on-the-job training
- **IAEA Technical Meeting on Status and Trends in Spent Fuel and Radioactive Waste Management** | Vienna, Austria | 28-31 Oct 2025
- **IAEA Regional Workshop on Developing, Enhancing and Implementing National Policy, Strategy and Programmes for Managing Spent Fuel and Radioactive Waste** | Vienna, Austria | 20-24 Jan 2025
- **IAEA Regional Workshop on Design Principles and Approaches for Radioactive Waste Repositories** | Peine, Germany | 4-8 Nov 2024
- **European Commission Tender Evaluation Committee - Technical Evaluator for TR4.01/23** | Brussels, Belgium | 29 Sep-11 Oct 2024
- **Additional IAEA / ENSTTI / WERC engagements in nuclear safety, management systems, construction management and radioactive-waste disposal concepts** | Europe and Asia | 2014-2019

EDUCATION

M.Sc. Candidate, Civil Engineering - Hacettepe University, Ankara | 2019-Present | Research focus: UAV-based radiation monitoring and mapping.

B.Sc., Civil Engineering - Middle East Technical University, Ankara | 2011 | Focus: Structural Engineering.

TECHNICAL SKILLS & LANGUAGES

Project & Program Delivery: scope and workflow definition, delivery roadmaps, interdependent workstreams, critical-path sequencing, milestones, dependencies, acceptance criteria, risk and issue management, stakeholder governance, procurement, vendor coordination and management reporting.

AI, Data & LLMs: OpenAI Playground, ChatGPT Pro, prompt and workflow design, Python, R, TensorFlow, Keras, PyTorch, pandas, NumPy, SciPy, SQL/PostgreSQL, data visualization, signal processing and geospatial analytics.

Applications & Integration: JavaScript/TypeScript, Flutter/Dart, C++, C#, web and mobile applications, cloud databases, Bluetooth/RF device integration and real-time sensor-data workflows.

Engineering & Governance: nuclear safety, radioactive waste management, safety case, regulatory licensing, management systems, QA/QC, SAP2000, SASSI, AutoCAD and Fusion 360.

Languages: Turkish (native); English (professional working proficiency).